Beyond GDP: The True Indicators of Extreme Poverty

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1 Abstract

Despite decades of global efforts to eradicate poverty, the gap between wealthy and impoverished nations continues to widen, exposing the limitations of traditional metrics like GDP in capturing the true state of human well-being. This paper seeks to move beyond GDP by analyzing the Multidimensional Poverty Index (MPI) as a more comprehensive indicator of extreme poverty. Using SAP's dataset, we investigate the underlying factors contributing to persistent poverty in countries with MPI values exceeding 0.15. By focusing on actionable insights, this study aims to equip national governments and international institutions, such as the World Bank and IMF, with strategies to foster sustainable development and alleviate extreme poverty effectively. Ultimately, our findings underscore the urgent need for new indicators and approaches that go beyond GDP to accurately reflect the lived experiences of the world's most vulnerable populations.

2 Introduction

In 1992, representatives from across the globe gathered at the United Nations, united by a bold vision: to eradicate poverty and declare October 17th the International Day for its elimination. Three decades later, the crisis is yet to be resolved.

It is easy to be fooled by rising GDP figures worldwide into assuming that conditions are improving for countries all over the world. However, they mask a harsh reality: wealth is accumulating at the top, while entire populations in the Global South continue to struggle with food insecurity, inadequate institutions, political instability, and lack of opportunity. The world's most impoverished nations have remained trapped in economic stagnation, with little benefit from the prosperity enjoyed by more developed countries. This growing divide exposes the failure of GDP as a true measure of progress and highlights the urgent need for new indicators that reveal the real state of extreme poverty.

This paper seeks to uncover why the world's poorest nations have continued to fail in alleviating poverty. In particular, it will analyze SAP's dataset to uncover the true indicators of extreme poverty as indicated by the Multidimensional Poverty Index (MPI). Based on these insights, the paper will propose targeted policy recommendations—both for national governments and international institutions like the World Bank and the IMF—to drive more effective and lasting solutions in the global fight against poverty. In particular, the following important questions are considered:

1. What key indicators that best capture different dimensions of poverty ?
2. Which indicators contribute most significantly to poverty levels ?
3. What policy interventions can be proposed to the most affected countries ?

In section 3, we describe the methodology used in data cleaning, feature selection, and model development. In section 4, the answers to each of the questions above are provided along with visualizations and summary statistics.

3 Methodology

3.1 Building a Multidimensional Index

The main objective of this task is to identify the features which correlate to the poverty indices to determine the main causes of poverty, and conceptualize policies that would target those features. There were three main problems that needed to be addressed: A significant amount of data was missing, which we will describe in more detail in 3.3. We decided to use mean substitution to fill in gaps in data for each country. There was two poverty metrics, one by UNDP and another by the World Bank. We decided to create two different models, one corresponding to each metric, and compare their results. Some of the data in the provided dataset were highly redundant and correlated. In the worst case, there were items with r squared greater

than 0.99. In our preliminary stages, we analyzed extremely high (r^2 greater than 0.80) intercorrelations and removed one item per pair. After addressing these initial hurdles, we performed the following procedure to create the index:

1. Run a regression model between all features and the poverty index.
2. Based on feature importance analysis, pick of the top 15 features.
3. Run 2 regression models with those new features.
4. Use those models to construct the equations for each.

3.2 UNDP MPI

During our initial exploration of the dataset, we noticed a problem: the majority of entries were missing for the Multidimensional Poverty Index (MPI). At first glance, it appeared that data was scarce—only 110 nation-year pairs were available for the UNDP's MPI indicator, while the World Bank's MPI indicator covered just 964 out of 6,360 possible pairs. However, upon closer examination, we realized the problem wasn't primarily due to missing data but rather because many countries reported an MPI value of 0. This revelation introduced a new challenge: any country with an MPI value of 0 was automatically deemed "perfect" by the index, regardless of its actual economic or social conditions. As a result, nations with vastly different standards of living and economic profiles, such as Singapore and Belarus, were both classified as having no poverty according to the MPI.

To gain a more accurate understanding of the indicators of extreme poverty and propose actionable policies, we decided to focus exclusively on countries with MPI values greater than 0. Specifically, this paper investigates the factors contributing to extreme poverty in countries where the MPI exceeds 0.15. Additionally, it explores potential policy measures that these nations could implement to reduce their MPI to below 0.15. The threshold of 0.15 was selected because it serves as a meaningful cutoff point around the median, as shown in 1.

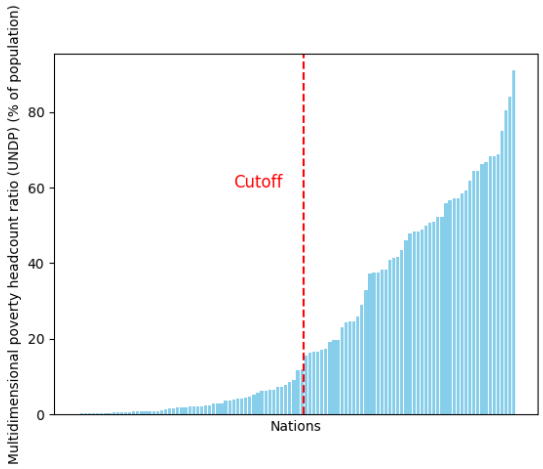


Figure 1: Visualization of cutoff

3.3 Data Cleaning

A significant portion of data was missing across multiple indicators in the dataset. Before analysis, we transformed the dataset from a wide to a long format, meaning that each row now represents a nation-year pair in this section. Out of 88 features, we found that just 3 had data for every nation-year row (2), with the median availability being less than 50 percent.

Since our approach focused on relationships between different features, missing data posed a significant challenge. For example, if we were using two features that were each available for less than 50% of all rows, we could expect that the number of direct row-to-row comparisons that could be made for the two features would be less than 50% of the total possible comparisons that could have been made if both datasets contained no missing values. As such, addressing these gaps was crucial in ensuring our analysis was meaningful.

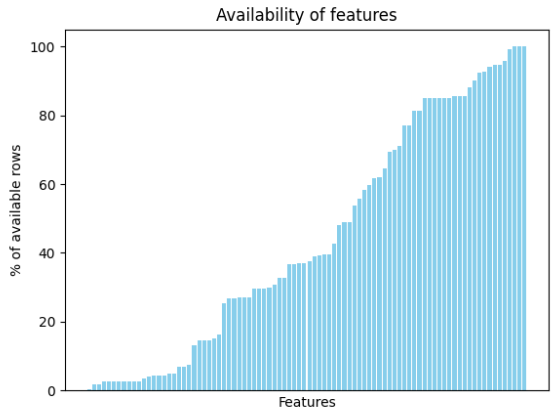
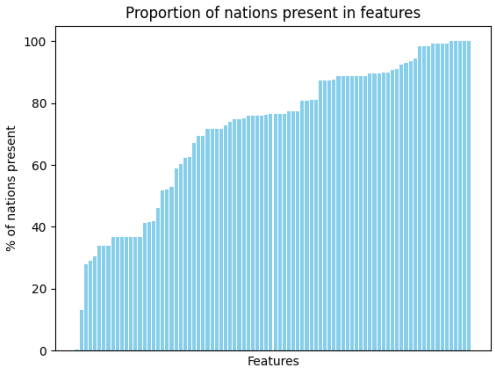
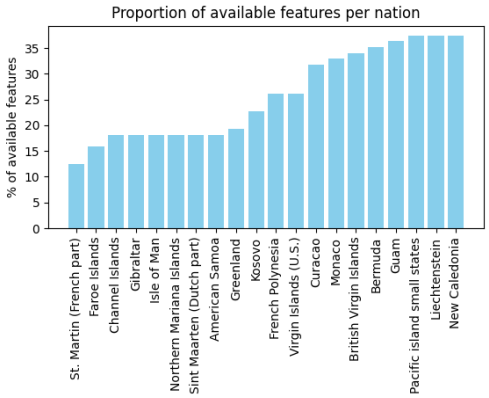


Figure 2: Total availability of features

Fortunately, we discovered that a large percentage of the missing values present in our long-format data was due to several indicators not reporting annually. When counting the number of nations present for each feature, we found that the median feature included at least one year row for 70 to 80 percent of nations, and that, ignoring one outlier, the worst features still included at least a quarter of the 265 nations provided (3a). As the dataset included constituent countries (e.g. Aruba) and world regions (e.g. Arab World) as nations, we realized that many rows that were missing from features were reporting on data from non-UN member nations (3b). We found that one way to alleviate the problem of missing years, taking into account



(a) Total availability of features



(b) Total availability of features

the presence of data across nations when years are ignored, was to leverage mean substitution. The median standard deviation of an indicator for one nation, across all years, was 2.22, so mean substitution across years was a convenient move that kept data mostly consistent.

3.4 Data Preprocessing

To ensure that the dataset was suitable for analysis, we underwent several preprocessing steps. First, redundant columns such as Country Code, short description, long description, Unit of measure, Topic, and Indicator Code were removed to focus on more relevant features. The data is then reshaped using the melt() function to unpivot year columns (e.g., "2000", "2001") into rows, converting it into a tidy format where each year represented a unique combination of Country Name, Indicator Name, Year, and Value. Another pivot operation using "pivot table()" restructured the data into a wide format, which listed each indicator as a column and rows indexed by Country Name and Year. Missing values were addressed by first sorting the data chronologically and then applying forward and backward filling within each country's time series to impute gaps under the assumption of temporal continuity. The final cleaned dataset was saved as "cleaned_dataset.csv", ready for our machine learning tasks.

3.5 Finding data-driven policies

A list of countries with MPI less than or equal to 15% in at least 4 out of the past 5 years (2018-2023) are deemed to be the most affected. The list consists of 108 out of 296 countries and territories represented in the original dataset. PCA is then conducted on this subset of data and the following indicators stand out the most:

1. Access to electricity (% of population) – PC1
2. Adjusted net national income per capita (current US$) – PC1
3. Unemployment, total (% of total labor force) – PC2
4. Multidimensional poverty headcount ratio (UNDP) – PC2
5. Multidimensional poverty headcount ratio (UNDP) – PC3
6. Unemployment, total (% of total labor force) – PC3
7. Access to electricity (% of population) – PC4
8. Adjusted net national income per capita (current US$) – PC4
9. Multidimensional poverty headcount ratio (UNDP) – PC1
10. Adjusted net national income per capita (current US$) – PC3
11. Adjusted net national income per capita (current US$) – PC2
12. Multidimensional poverty headcount ratio (UNDP) – PC4
13. Access to electricity (% of population) – PC2
14. Unemployment, total (% of total labor force) – PC4
15. Unemployment, total (% of total labor force) – PC1
16. Access to electricity (% of population) – PC3

3.6 Feature Selection/Data Engineer

After conducting exploratory data analysis and PCA, the following variables are included in our final training and testing datasets:

1. Urban population (% of total population)
2. Multilateral debt service (% of public and publicly guaranteed debt service)

3. Adjusted savings: education expenditure (current US$)
4. Political Stability and Absence of Violence/Terrorism: Estimate
5. Literacy rate, adult total (% of people ages 15 and above)
6. Access to electricity (% of population)

Finally, the typical approach in normalizing numeric values is used in our analysis, namely

$$\mathbf{x} \mapsto \frac{\mathbf{x} - \mu}{\sigma},$$

where μ is the mean and σ is the standard deviation of $\mathbf{x}$.

3.7 Training, Validating, and Testing Sets

After performing data cleaning, feature selection, and data engineering, we split the dataset into training set (60%), validating set (20%), and testing set (20%) using shuffling and random sampling.

The United Nations uses the Human Development Index (HDI) as a summary measure for assessing long-term progress in life expectancy, education, and economic health. It provides a useful view of a country's economic and social development. To test our model, we benchmarked against this index.

3.8 Prediction Model

The model that we pick in this case is Extreme Gradient Boosting Regressor (XGBoost Regressor) [**?**]. We find that this model has a nice balance between interpretability and predictive power. Furthermore, the model provides feature importance ranking for the independent variables as well as great performance on structured data.

4 Experiments and Results

4.1 Multidimensional Index

From 3.1, we determined that the models based on UNDP produced better results. The model obtained was the following:

Poverty Index = .0043 Adequacy of social protection and labor programs (% of total welfare of beneficiary households)
+.0047 Age dependency ratio, old (% of working-age population)
+.0059 Access to clean fuels and technologies for cooking (% of population)
−.0073 Age dependency ratio (% of working-age population)
−.0081 Access to electricity (% of population)
+.0111 Adjusted net national income per capita (current US$)
−.0182 Adequacy of unemployment benefits and ALMP (% of total welfare of beneficiary households)
+.0259 Adolescents out of school, male (% of male lower secondary school age)
+.0259 Adjusted savings: education expenditure (current US$)
−.0320 Adequacy of social insurance programs (% of total welfare of beneficiary households)
−.0431 Adolescents out of school, female (% of female lower secondary school age)
−.1249 Adolescent fertility rate (births per 1,000 women ages 15-19)
+.1574 Adolescents out of school (% of lower secondary school age)
−.2172 Adequacy of social safety net programs (% of total welfare of beneficiary households)
−.3141 Age dependency ratio, young (% of working-age population)

4.2 Targeted Poverty Alleviation Policies

From 3.5, we determined certain metrics that needed to be focused on. The main metrics include access to electricity, per capital national income, addressing unemployment and improving overall public health via increasing access to healthcare, clean water, and sanitation.

4.2.1 Expanding Access to Electricity (PC1, PC4)

One of the most effective ways to combat poverty is to ensure that all individuals, especially in rural and underdeveloped areas, have access to reliable electricity. Governments and international organizations should prioritize electrification through affordable and sustainable means, such as renewable energy sources like solar power, wind turbines, and hydropower. Decentralized solutions, including solar microgrids, can be a cost-effective approach, particularly in areas that are not connected to the main power grid. Furthermore, these systems can be scaled to meet the needs of the population over time.

The relationship between access to electricity and poverty reduction is well-established. Electricity can improve quality of life by enabling people to access essential services, such as education, healthcare, and financial services. For example, electrification can increase study hours for students, improve healthcare outcomes through refrigerated medicines, and support entrepreneurship through energy access. According to Banerjee, Duflo, and Qian (2015), access to electricity leads to higher productivity, income generation, and employment opportunities, which directly impacts poverty levels.

As a case study, the World Bank's Lighting Africa initiative has successfully expanded electricity access to millions of people in Sub-Saharan Africa by providing affordable solar energy products. The initiative not only contributes to reducing poverty but also mitigates environmental damage by replacing kerosene lamps with clean, renewable energy sources.

4.2.2 Increasing National Income Per Capita (PC1, PC4)

Governments in low-income countries can work toward increasing national income per capita through targeted economic reforms, such as progressive taxation systems, boosting local entrepreneurship, and diversifying economies. By ensuring that the benefits of growth are distributed fairly, governments can stimulate economic development while minimizing income inequality. Additionally, governments can invest in industries such as agriculture, tourism, and technology to create jobs and diversify the sources of income in the economy.

Higher national income is a strong indicator of a country's overall economic well-being. As national income rises, there is generally a reduction in poverty rates and an improvement in living standards. The World Bank (2021) emphasizes that economic growth coupled with income redistribution policies is essential for achieving sustainable poverty reduction. Countries that have invested in creating robust economies have seen significant decreases in poverty, while improving access to social services, infrastructure, and technology.

4.2.3 Decreasing Unemployment (PC2, PC3)

Unemployment, particularly in developing economies, is a key factor driving poverty. Governments should focus on programs that foster job creation, particularly in sectors such as agriculture, manufacturing, and services. Investments in vocational education and skill development programs are crucial, as they help workers gain marketable skills that align with industry demands. Additionally, creating microfinance opportunities and supporting the informal sector can help stimulate entrepreneurship and job creation.

Unemployment is a major driver of poverty, especially in developing countries where labor markets are often underdeveloped, and opportunities for formal employment are limited. The International Labour Organization (2022) stresses that providing accessible job training and economic opportunities can significantly reduce poverty by increasing the employability of the labor force and decreasing the dependency on government assistance.

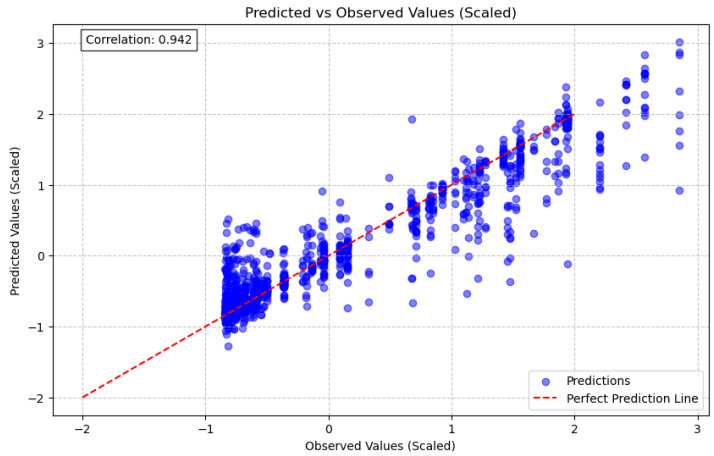
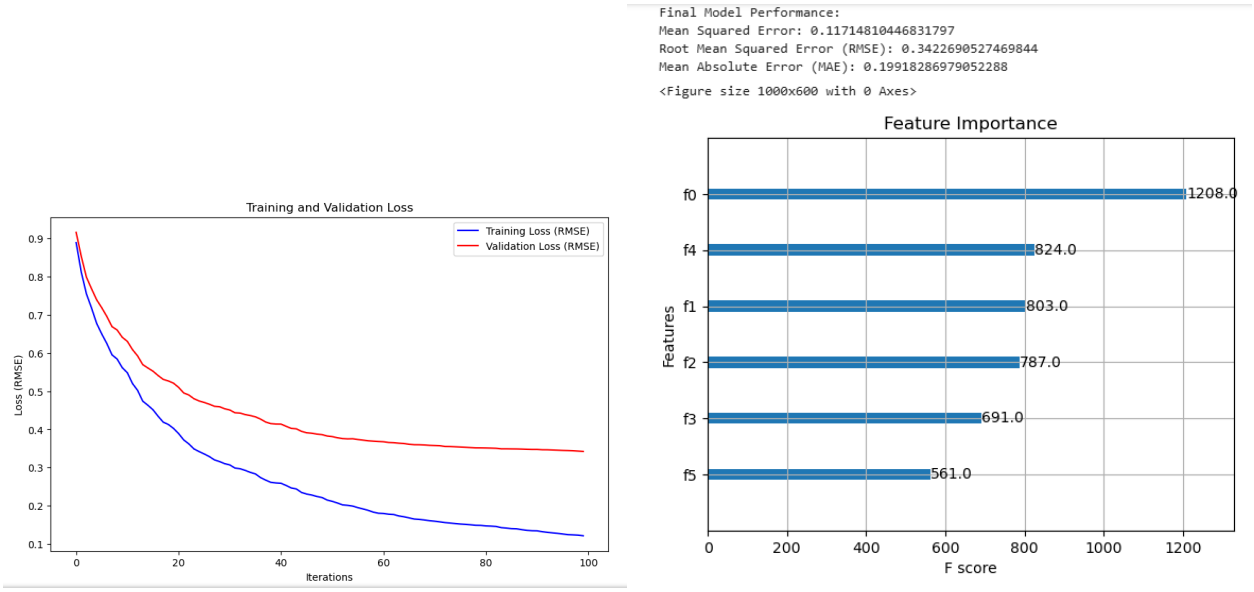


Figure 4: Test Set vs Predictions



(a) Training and Validating Losses

(b) Feature Importance

Figure 5: Losses and Feature Importance

5 Conclusion

In conclusion, this study underscores the critical need to move beyond traditional metrics like GDP in understanding and addressing the complexities of extreme poverty. By analyzing the Multidimensional Poverty Index (MPI) and leveraging advanced machine learning techniques such as XGBoost , we have developed a robust index that captures the multidimensional nature of deprivation more accurately than conventional measures. Our findings reveal that factors such as access to electricity, unemployment rates, national income per capita, and social safety nets are key contributors to poverty levels, particularly in countries with MPI values exceeding 0.15. These insights provide a clear roadmap for targeted policy interventions aimed at fostering sustainable development and improving the lives of the world's most vulnerable populations.

Through rigorous data cleaning, feature selection, and cross-validation with benchmarks like the Human Development Index (HDI), we ensured the reliability and relevance of our results. The creation of an in-

dex that aligns well with established indicators highlights the potential of data-driven approaches to inform evidence-based policymaking. Furthermore, our analysis emphasizes the importance of addressing systemic issues such as inadequate infrastructure, economic inequality, and limited access to essential services like healthcare and education.

Looking ahead, the global fight against poverty requires a paradigm shift toward more comprehensive and nuanced measures of human well-being. By adopting multidimensional frameworks like the one proposed in this study, national governments and international institutions can design and implement policies that not only reduce poverty but also promote long-term resilience and equity. As the world continues to grapple with persistent inequalities, this research serves as a call to action for stakeholders to prioritize innovative solutions that reflect the lived realities of those most affected by extreme poverty. Ultimately, our work demonstrates that progress is possible—but only if we are willing to look beyond outdated metrics and embrace new ways of measuring and addressing the challenges of our time.